

MIRUNALINI A

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PROFILE

Aspiring Software/Data Engineer pursuing B.Tech in Electronics and Computer Engineering at VIT Chennai (CGPA: 8.59), with hands-on experience building full-stack AI-powered products, predictive/analytics models, and embedded systems. Comfortable across the stack — from system design and architecture to data-driven modeling — with demonstrated, responsible use of AI (Gemini, RAG, multi-agent pipelines) in production systems.

EDUCATION

Vellore Institute of Technology, Chennai B.Tech – Electronics and Computer Engineering | CGPA: 8.59 2023 – 2027

Everwin Vidhyashram (CBSE) Class 12 – CS Stream: 91% | Class 10: 85% 2019 – 2023

WORK EXPERIENCE

Ikomet Technologies AI & Software Engineering Intern Jun 2026 – Jul 2026

- Contributed to a RAG-based SAT Question Generation System using Generator–Validator AI agents; performed QA validation of AI-generated content to improve system accuracy and reliability.
- Reviewed AI-driven learning platform content and UI/UX, and researched curriculum and localization requirements to support product expansion.

Qmax Test Equipments Pvt Ltd Embedded & IC Testing Intern May 2025 – Jun 2025

- Used Automated Test Equipment (ATE) for IC model programming and AC/DC parametric testing; automated test routines using Python and IEEE VHDL.
- Interpreted IC datasheets and applied graphical learn-and-compare techniques for functional verification and diagnostic analysis.

Heavy Vehicles Factory, Avadi, Chennai Inplant Trainee Dec 2024 – Jan 2025

- Gained exposure to system validation, parts testing, and assembly-line workflows in a large-scale manufacturing environment.

PROJECTS

AI-Powered SAT Question Generation System — React, TypeScript, Express.js, Gemini AI, Qdrant, RAG

- Built a full-stack AI platform automating SAT question generation using a Generator–Validator multi-agent architecture; integrated Qdrant vector DB with semantic embeddings for context-aware RAG, improving accuracy and consistency.

Collaborative Spreadsheet Application — Next.js, TypeScript, Firebase, Firestore, Realtime DB

- Built a real-time collaborative spreadsheet with live presence tracking and conflict resolution; engineered a custom spreadsheet engine (formulas, CSV/XLSX export) with optimistic UI and debounced real-time sync.

AI-Enabled Wastewater Treatment Predictive Control System — Transformer Neural Network, ONNX, MATLAB/Simulink, MPC

- Trained a Transformer model to forecast wastewater treatment parameters, exported to ONNX, and integrated into Simulink to drive a Model Predictive Controller — an end-to-end analytics-to-control pipeline.

AgroBot — ML-Powered Precision Agriculture Robot — Raspberry Pi 5, Python, ML, Computer Vision, Mobile App

- Built a CV-based plant disease classification pipeline on Raspberry Pi 5 triggering precision-targeted pesticide spraying to reduce chemical wastage.
- Built a companion mobile app for live robot monitoring, control, and field analytics; selected for Smart India Hackathon 2025 (SIH'25). Patent published for the detection and spraying mechanism.

Remote Health Monitoring System — ESP32, Embedded C, WiFi, Firebase, React Native

- Designed a real-time patient monitoring system integrating multiple physiological sensors on a custom PCB, with ESP32 firmware for continuous multi-sensor acquisition and interrupt-driven real-time operation.
- Built a companion mobile app and web dashboard for live visualization, alert management, and remote monitoring over WiFi-based cloud communication.

PATENTS

- Closed-Loop Liquid Dispensing System with Image-Based Spatial Targeting and Deposition Verification — Published (App. No. 202641065168, Pub. No. 23/2026).
- Maternal-Fetal Health Monitoring and Risk Prediction System — Filed; edge-AI real-time prediction of fetal distress using wireless sensor fusion.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript

Full-Stack & Mobile: React, Next.js, Node.js, Express.js, HTML, CSS, Tailwind CSS, React Native, Flask, Firebase, Firestore, Realtime Database

AI/ML & GenAI: Machine Learning, TensorFlow Lite, Gemini AI, RAG, Qdrant (Vector DB), NumPy, pandas, scikit-learn, Signal Processing, MATLAB/Simulink

Embedded & IoT: ESP32, Arduino, Embedded C, VHDL, Verilog, PCB Design, LTspice

Tools & Fundamentals: Git, Jupyter Notebook, OOPS, DSA, Operating Systems, Database Concepts

CERTIFICATIONS & EXTRA-CURRICULAR

Spoken Tutorial (IIT Bombay) – ns-3 Training | EduPyramids (SINE IIT Bombay) – Java & Advanced C++ Training | WEBDEX Hackathon – ACM SC VITC | Vega Processors Workshop, VIT Chennai | Udemy – PCB Design with Eagle | Marketing Member, CodeChef Club VIT Chennai

LANGUAGES KNOWN

English (Fluent) | Tamil (Fluent) | French (Basics)